

Review Chapter 1

Applied Algebra
Instructor: Tuan Tran

The division algorithm and greatest common divisor

- ▶ division with remainder;
- ▶ the Euclidean algorithm;
- ▶ greatest common divisors and Bezout's identity.

Division algorithm

Let $a, b \in \mathbb{Z}$ with $a > 0$.

There exist integers q, r such that

$$b = aq + r, \quad 0 \leq r < a.$$

- ▶ q is the quotient;
- ▶ r is the remainder.

This decomposition is unique.

Example (positive numbers)

Write 175 in the form

$$175 = 11q + r, \quad 0 \leq r < 11.$$

We find:

$$175 = 11 \cdot 15 + 10.$$

So

$$q = 15, \quad r = 10.$$

Example (negative numbers)

Write -47 in the form

$$-47 = 6q + r, \quad 0 \leq r < 6.$$

We find:

$$-47 = 6(-8) + 1.$$

Important. The remainder must be non-negative.

Divisibility

Definition

We say that a divides b , written

$$a \mid b,$$

if

$$b = ak$$

for some integer k .

Examples.

$$7 \mid 42, \quad 7 \nmid 40.$$

Every integer divides 0.

Greatest common divisor

Definition

The greatest common divisor of a and b , denoted $\gcd(a, b)$, is the largest integer d such that

$$d \mid a \quad \text{and} \quad d \mid b.$$

Examples.

$$\gcd(12, 30) = 6, \quad \gcd(4, 15) = 1.$$

Euclidean algorithm

Key idea.

$$\gcd(a, b) = \gcd(a, r),$$

where r is the remainder when dividing b by a .

We repeatedly reduce the problem:

$$(a, b) \rightarrow (a, r) \rightarrow (r, r') \rightarrow \dots$$

Each step produces smaller numbers.

We repeat this process until the remainder is 0. The last non-zero remainder is the gcd.

Example (Euclidean algorithm)

Compute $\gcd(252, 105)$.

$$252 = 2 \cdot 105 + 42$$

$$105 = 2 \cdot 42 + 21$$

$$42 = 2 \cdot 21$$

$$\gcd(252, 105) = 21.$$

Linear combination (Bézout identity)

There exist integers s, t such that

$$\gcd(a, b) = as + bt.$$

This expresses the gcd as a combination of a and b .

Example (linear combination)

Find s, t such that

$$21 = 252s + 105t.$$

$$\begin{aligned} 21 &= 105 - 2 \cdot 42 \\ &= 105 - 2(252 - 2 \cdot 105) \\ &= 5 \cdot 105 - 2 \cdot 252 \end{aligned}$$

So

$$21 = -2 \cdot 252 + 5 \cdot 105.$$

Coprime integers

Definition

We say a and b are coprime if

$$\gcd(a, b) = 1.$$

Example.

$$\gcd(15, 8) = 1.$$

Key fact. If $\gcd(a, b) = 1$ and $a \mid bc$, then

$$a \mid c.$$

Prime Factorization and Proof by Induction

- ▶ prime numbers;
- ▶ prime factorization;
- ▶ mathematical induction.

Prime numbers

Definition

A positive integer $n > 1$ is **prime** if its only positive divisors are
1 and n .

Otherwise, it is called **composite**.

Examples.

2, 3, 5, 7, 11 are prime, $6 = 2 \cdot 3$ is composite.

Examples

Which of the following are prime?

13, 21, 29, 35

Answer.

- ▶ 13 is prime;
- ▶ $21 = 3 \cdot 7$ is composite;
- ▶ 29 is prime;
- ▶ $35 = 5 \cdot 7$ is composite.

Why primes matter

- ▶ Every integer can be built from primes;
- ▶ understanding primes helps us understand all integers.

Prime factorization

Theorem

Every integer greater than 1 can be written as a product of primes. This factorization is unique (up to order).

Example.

$$60 = 2^2 \cdot 3 \cdot 5.$$

$$840 = 84 \cdot 10 = (2^2 \cdot 3 \cdot 7)(2 \cdot 5) = 2^3 \cdot 3 \cdot 5 \cdot 7.$$

Gcd via prime factorizations

Let $a = 180$ and $b = 72$. Then

$$a = 2^2 \cdot 3^2 \cdot 5, \quad 72 = 2^3 \cdot 3^2.$$

Hence

$$\gcd(a, b) = 2^2 \cdot 3^2 = 36.$$

The gcd takes the smallest exponent of each prime.

Another example

Let $a = 120$ and $b = 252$. Then

$$a = 2^3 \cdot 3 \cdot 5, \quad b = 2^2 \cdot 3^2 \cdot 7.$$

Hence

$$\gcd(a, b) = 2^2 \cdot 3 = 12.$$

Rule. Take the minimum exponent of each common prime.

Least common multiple

Definition

The least common multiple of a and b , denoted $\text{lcm}(a, b)$, is the least integer $n \geq 1$ such that

$$a \mid n \quad \text{and} \quad b \mid n.$$

Example.

$$108 = 2^2 \cdot 3^3, \quad 72 = 2^3 \cdot 3^2.$$

$$\text{lcm}(108, 72) = 2^3 \cdot 3^3 = 216.$$

Rule. Take the largest exponent of each prime.

Relation between gcd and lcm

For positive integers a, b :

$$\gcd(a, b) \cdot \text{lcm}(a, b) = ab.$$

Example.

$$\gcd(108, 72) = 36, \quad \text{lcm}(108, 72) = 216.$$

$$\gcd(108, 72) \cdot \text{lcm}(108, 72) = 36 \cdot 216 = 108 \cdot 72.$$

Induction (idea only)

We often want to prove statements for all integers n .

Method:

- ▶ check $n = 1$;
- ▶ assume true for n ;
- ▶ prove for $n + 1$.

This is called **mathematical induction**.

Example (induction)

Claim:

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Check $n = 1$:

$$1 = \frac{1 \cdot 2}{2}.$$

Assume true for n , then for $n + 1$:

$$1 + \cdots + n + (n + 1) = \frac{n(n+1)}{2} + (n + 1) = \frac{(n+1)(n+2)}{2}.$$

Congruence classes

- ▶ congruence modulo n ;
- ▶ addition and multiplication;
- ▶ invertible congruence classes;
- ▶ solving equations;
- ▶ applications.

Congruence

Definition.

We say

$$a \equiv b \pmod{n}$$

if

$$n \mid (a - b).$$

That is, $a \equiv b \pmod{n}$ if and only if a and b have the same remainder when divided by n .

Examples:

$$15 \equiv 1, \quad -6 \equiv 1, \quad 22 \equiv 1 \pmod{7}.$$

Congruence classes

For fixed n , define

$$[a]_n = \{a + kn : k \in \mathbb{Z}\}.$$

Example (mod 5):

$$[0]_5, [1]_5, [2]_5, [3]_5, [4]_5.$$

These sets partition the integers.

Basic properties

If

$$a \equiv b \pmod{n}, \quad c \equiv d \pmod{n},$$

then

$$a + c \equiv b + d \pmod{n},$$

$$ac \equiv bd \pmod{n}.$$

Arithmetic on $\mathbb{Z}_n = \{[0]_n, [1]_n, \dots, [n-1]_n\}$.

$$[a]_n + [b]_n = [a + b]_n,$$

$$[a]_n [b]_n = [ab]_n.$$

Example (computation)

Compute:

$$3^{20} \bmod 7.$$

$$3^2 \equiv 2, \quad 3^4 \equiv 4$$

$$3^8 \equiv 2, \quad 3^{16} \equiv 4$$

$$3^{20} = 3^{16} \cdot 3^4 \equiv 4 \cdot 4 = 16 \equiv 2.$$

Invertible congruence classes

Definition

A class $[a]_n$ is **invertible** if there exists $[b]_n$ such that

$$[a]_n[b]_n = [1]_n.$$

In that case, $[b]_n$ is the inverse of $[a]_n$ and is written $[a]_n^{-1}$.

Theorem. $[a]_n$ is invertible if and only if $\gcd(a, n) = 1$.

Cancelation law. Suppose $\gcd(a, n) = 1$. Then $ac \equiv ab \pmod{n}$ implies $b \equiv c \pmod{n}$.

Solving $ax \equiv b \pmod{n}$

- 1 Compute $d = \gcd(a, n)$.
- 2 If $d \nmid b$: no solution. STOP.
- 3 If $d \mid b$, divide through by d to get

$$(a/d)x \equiv b/d \pmod{n/d}.$$

- 4 Since $\gcd(a/d, n/d) = 1$, find the inverse of a/d modulo n/d .
- 5 Multiply by that inverse to get one solution c modulo n/d .
- 6 Lift to all d solutions modulo n :

$$c, c + n/d, \dots, c + (d - 1)n/d.$$

Example (solving)

Solve

$$9x \equiv 3 \pmod{12}.$$

$$\gcd(9, 12) = 3 \mid 3 \Rightarrow \text{solutions exist.}$$

Divide:

$$3x \equiv 1 \pmod{4}.$$

$$x \equiv 3 \pmod{4}.$$

Chinese Remainder Theorem

Suppose we have:

$$x \equiv a \pmod{m}, \quad x \equiv b \pmod{n},$$

where $\gcd(m, n) = 1$.

Theorem. There exists a unique solution modulo mn .

Example (Chinese remainder theorem)

Solve:

$$x \equiv 1 \pmod{3}, \quad x \equiv 2 \pmod{5}.$$

Since $x \equiv 1 \pmod{3}$, we can write $x = 3k + 1$ for some integer k .

The equation $x \equiv 2 \pmod{5}$ becomes

$$3k \equiv 1 \pmod{5}.$$

We get $k \equiv 2 \pmod{5}$, that is,

$$k = 5l + 2 \quad \text{for some integer } l.$$

So $x = 3k + 1 = 15l + 7$. We find $x \equiv 7 \pmod{15}$.

Euler's theorem

Define $\varphi(n)$ as the number of integers $1 \leq k \leq n$ such that $\gcd(k, n) = 1$.

Example. $\varphi(10) = 4$ (1, 3, 7, 9).

Theorem. If $(a, n) = 1$, then

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Formula. If $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ is the prime factorization of n , then

$$\varphi(n) = n \left(1 - \frac{1}{p_1}\right) \cdots \left(1 - \frac{1}{p_r}\right) = p_1^{\alpha_1-1}(p_1 - 1) \cdots p_r^{\alpha_r-1}(p_r - 1).$$

Example.

$$\varphi(10) = 10 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right) = 4.$$